

PROJECT: DATA ROOM UX & FLEET



Data Room UX & Fleet Management

Version: v0.5.10 | Date: 21 Feb 2026

Developer Brief: A secure, ephemeral, browser-based environment for sensitive data sharing.

FROM: Human (Project Lead)
TO: Developer (Lead), Designer, DevOps

THE SUPER-DOCUMENT MODEL

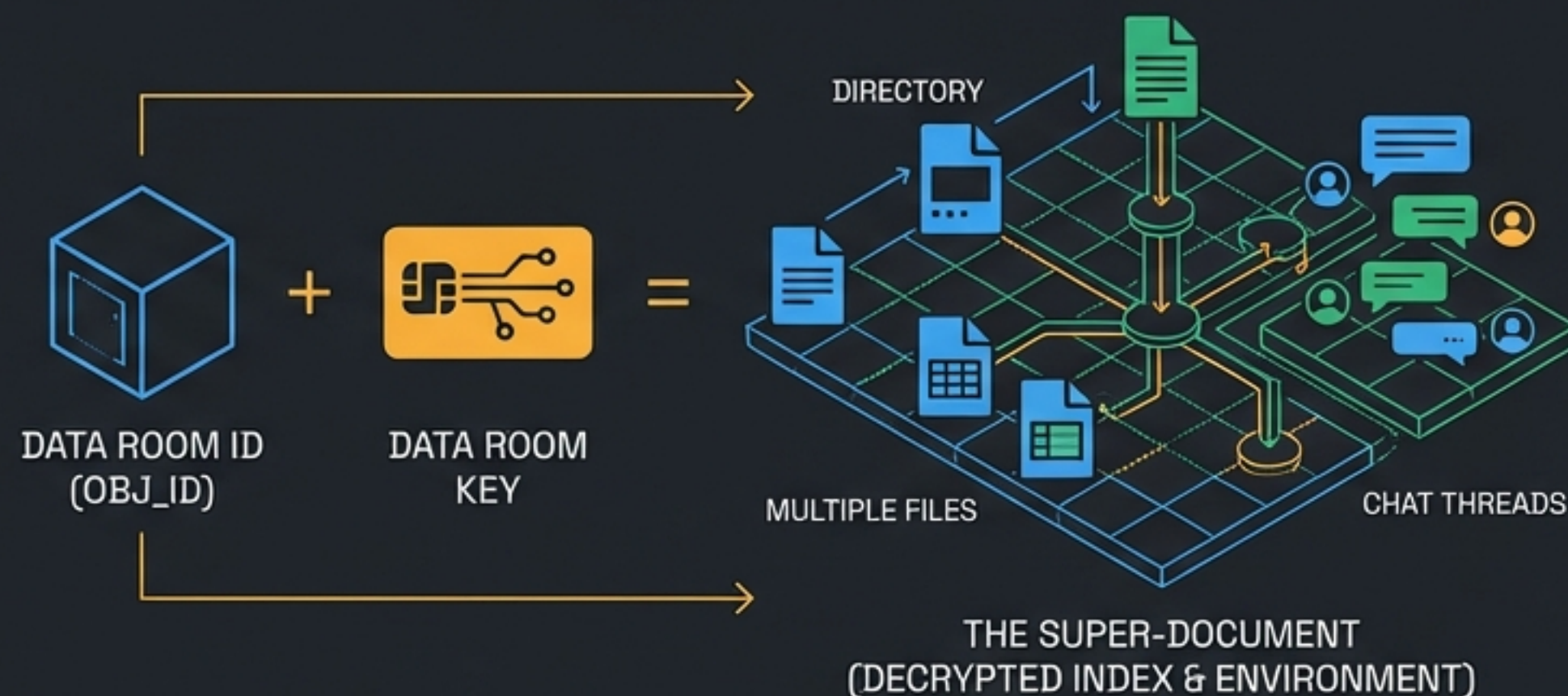
The key no longer just opens a file; it unlocks the index to an entire room.

CURRENT MODEL



`decrypt(doc_id, sym_key) -> file_data`

DATA ROOM MODEL



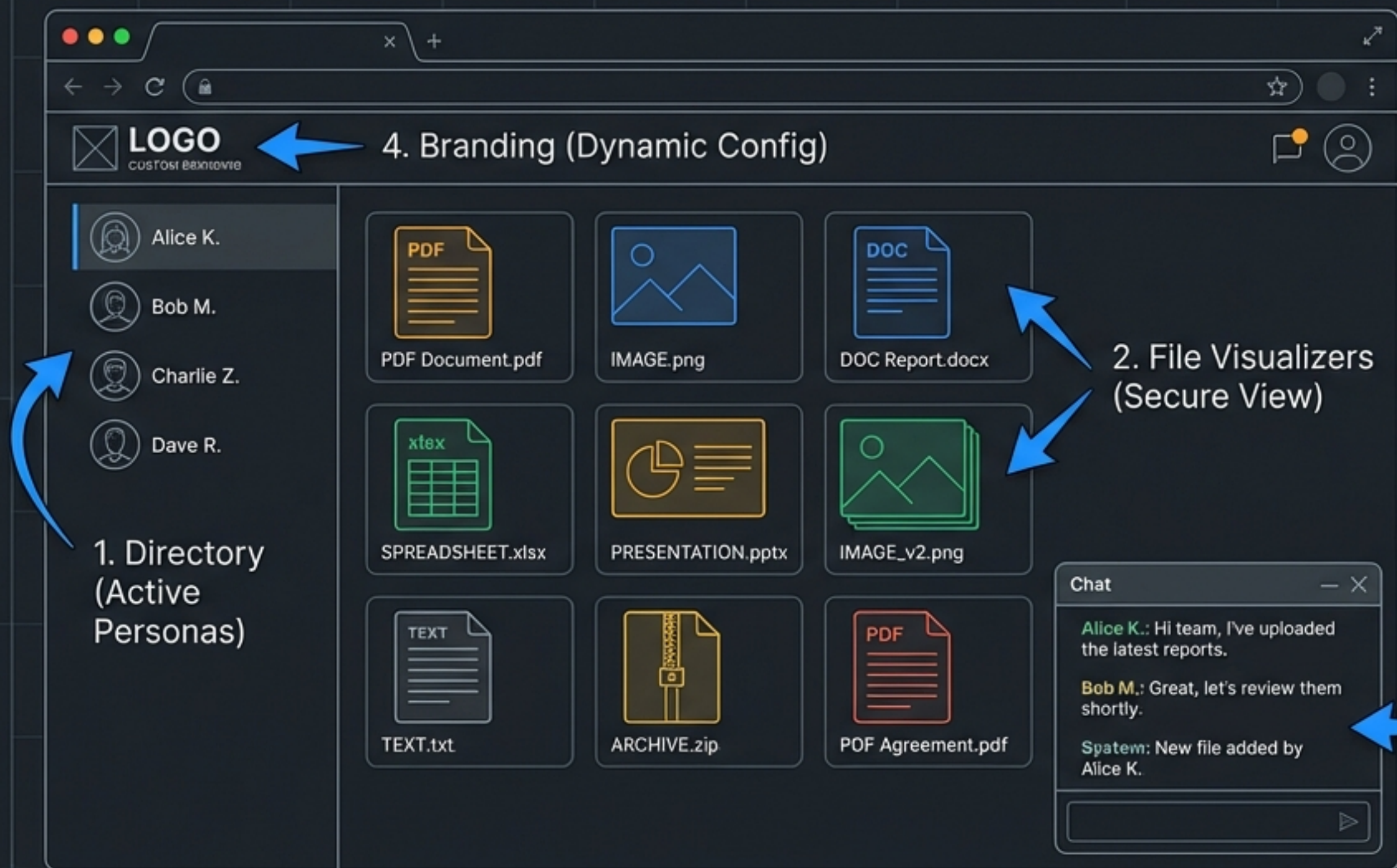
`decrypt(obj_id, room_key) -> {directory, files[], chat_history[]}`

CONTENT ID: Identifies the specific Data Room (Obj_Id).

DECRYPTION KEY: A symmetric key shared via a second channel.

SCOPE: Decrypts the room structure, directory, and file references.

THE USER EXPERIENCE: INSIDE THE ROOM



Entry: Once unlocked via the entry URL (e.g., `investor-x.send.sgraph.ai`), the user enters a branded environment served directly from the EC2 instance.

Config: Custom branding (logo, colours) is rendered dynamically from the room configuration.

3. Messages (Encrypted Overlay)

SECURE VISUALIZATION (NO DOWNLOADS)

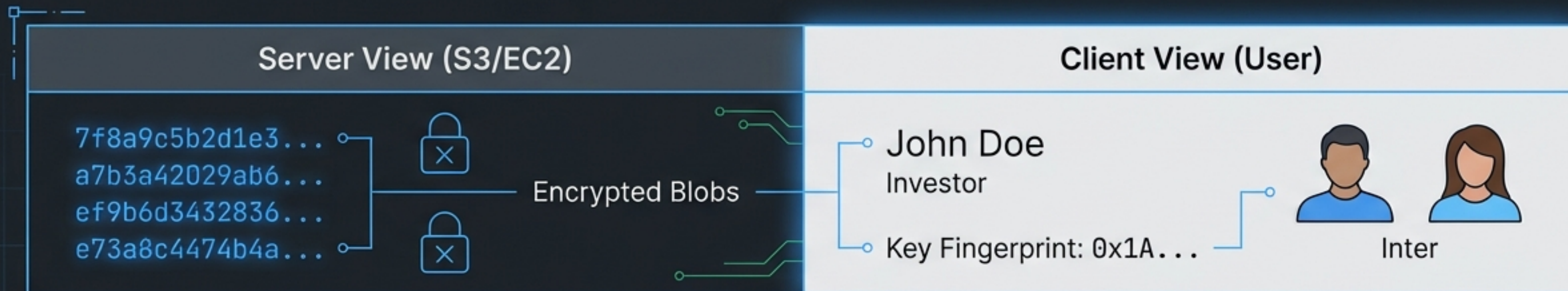
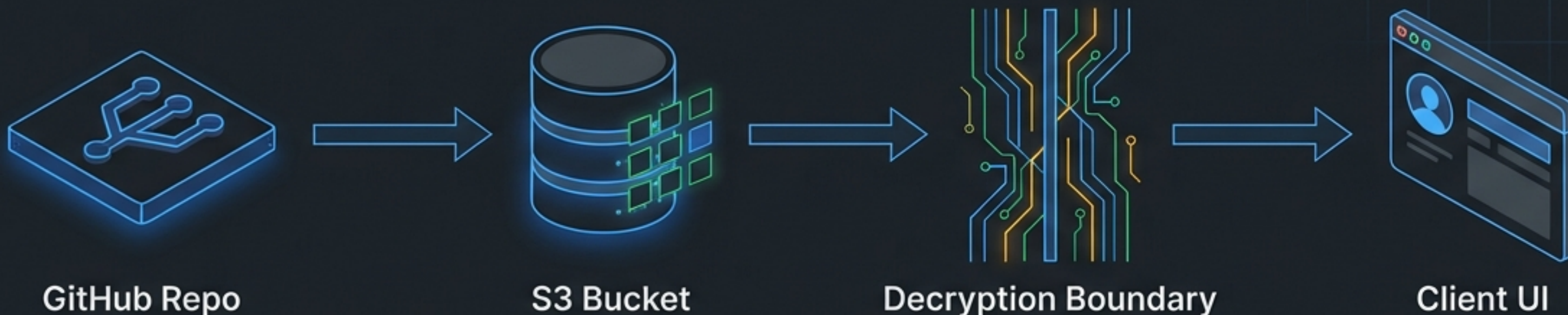
The Sandbox Constraint: All rendering occurs in sandboxed iframes (`sandbox='allow-same-origin'``).



Unsupported files offer download only. No inline previews for unknown types.

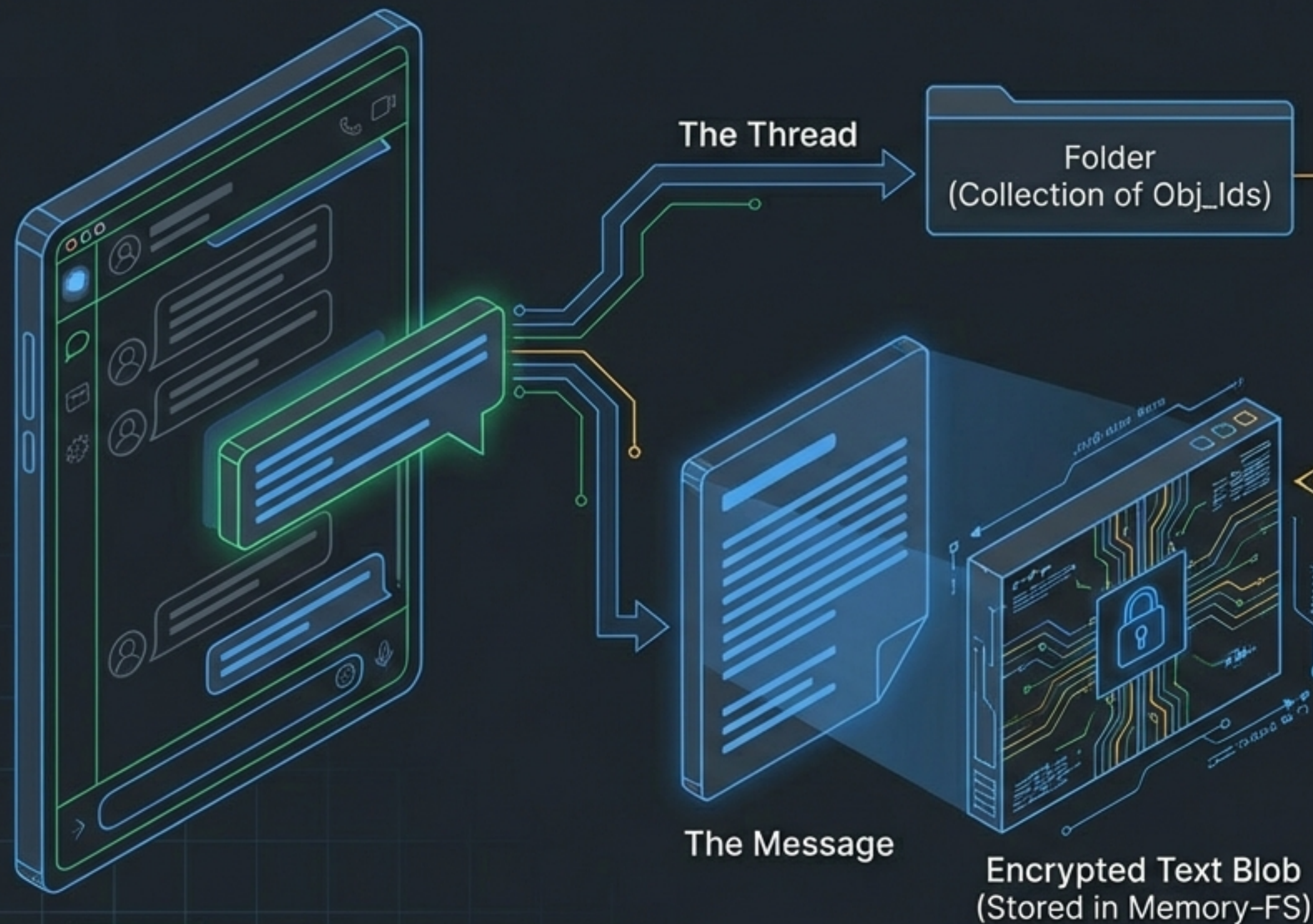
DIRECTORY & IDENTITY

The server hosts the party but doesn't know the guests.



Directory data originates in GitHub, syncs to S3, and is decrypted only by the client holding the Data Room Key.

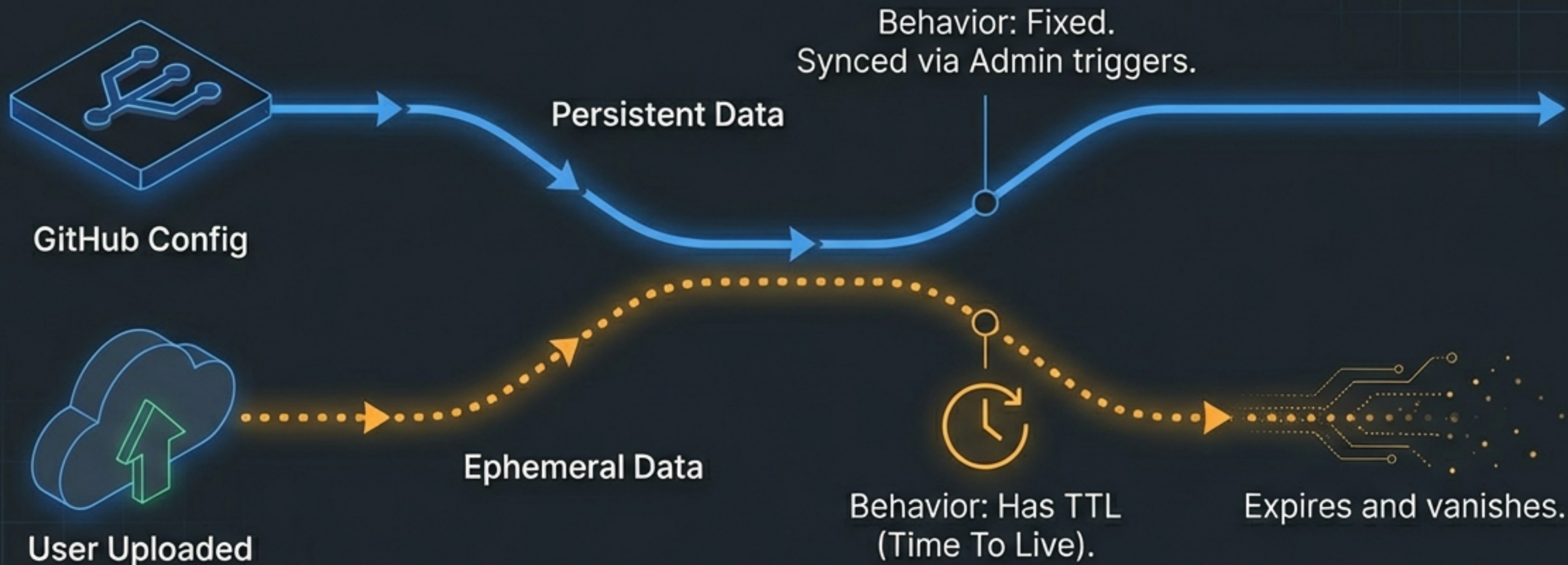
ENCRYPTED MESSAGING



PKI-Encrypted Architecture

-  **Encryption:** Sender encrypts to the Recipient's public key.
-  **Structure:** Messages are treated as encrypted text files.
-  **Structure:** Messages are treated as encrypted text files.
-  **Privacy:** The server stores the sequence for ordering but cannot read the content (timestamp, sender, attachments are all inside the encrypted blob).

STORAGE TIERS: PERSISTENT VS. EPHEMERAL



UX Implication: The UI must visually distinguish these tiers. Users must understand that ephemeral files will expire.

FLEET MANAGEMENT (ADMIN)

API-First. The UI is merely a convenience layer for the Fleet Management API.

KEY CAPABILITIES



Monitor: View status of all rooms.



Control: Start (trigger EC2 boot) / Stop (sync & terminate).



Sync: Trigger GitHub → S3 updates.

Mission Control /admin/fleet.html

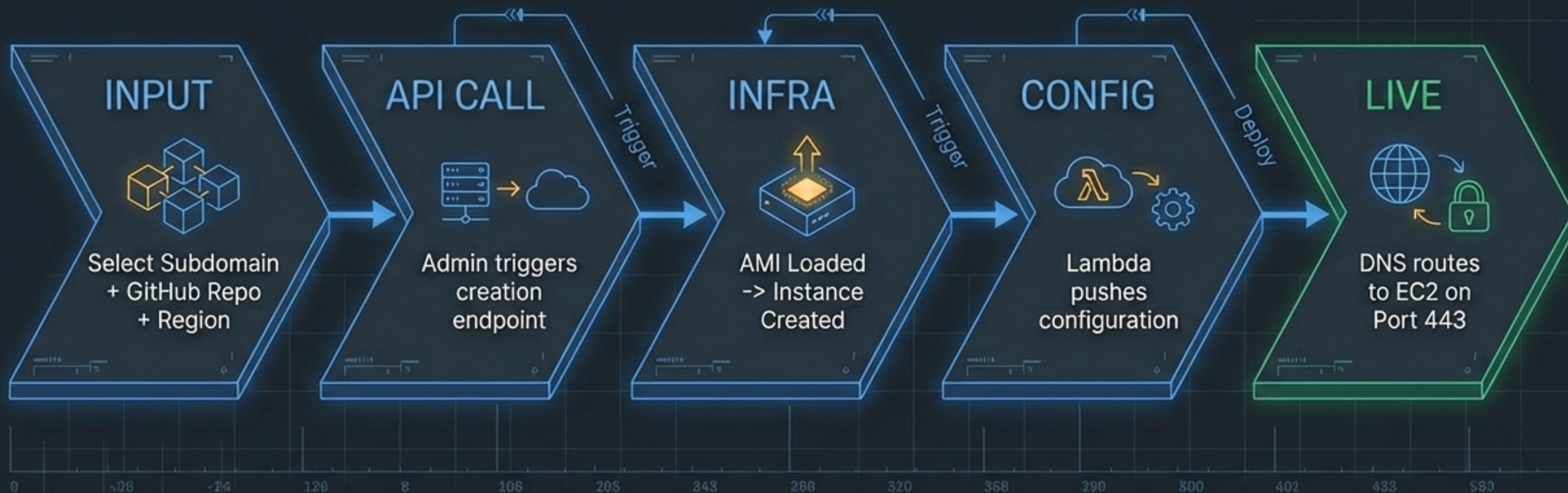
	ROOM ID	STATUS	ACTION
1	investor-alpha	● Running	☐ Stop
2	deal-bravo	● Stopped	☐ Start
3	audit-charlie	● Booting	☐ Loading... 
			☐ Execute



UX Implication: Administrative actions have immediate, tangible impact on resources. Actions should be clear and confirmable.

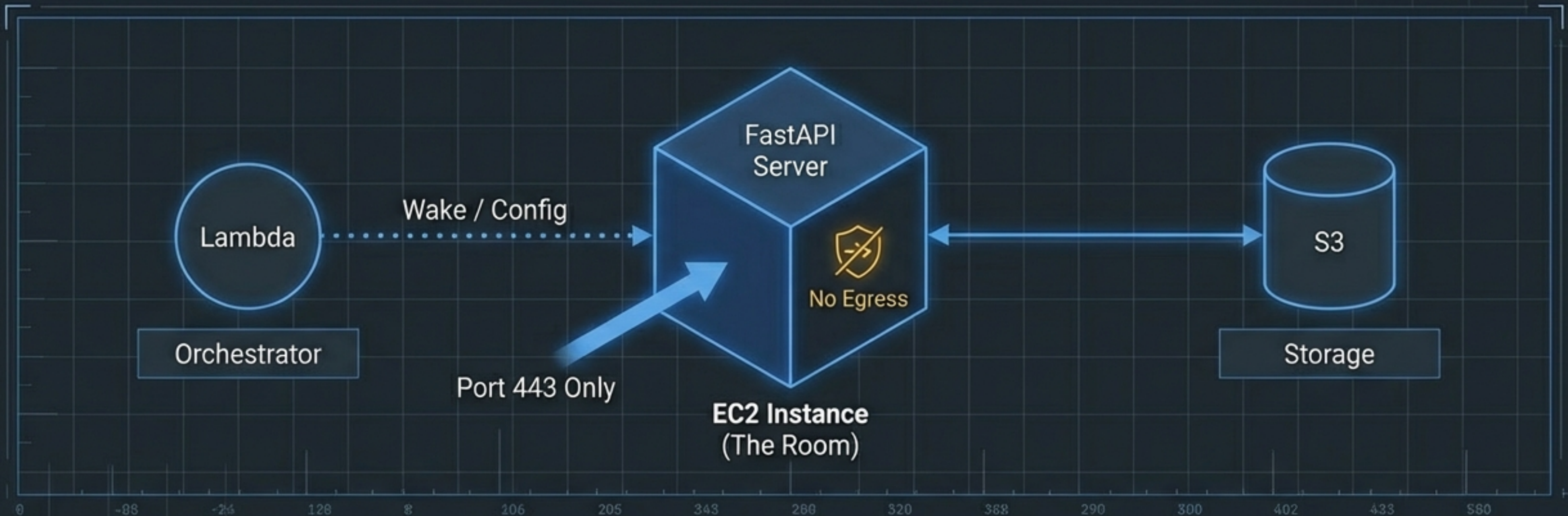
THE CREATION WORKFLOW

Linking a GitHub repository configuration to a transient EC2 resource.



DEVOPS INFRASTRUCTURE: THE ISOLATED INSTANCE

An isolated, ephemeral environment for secure application hosting.



Lifecycle: Instance shuts down after idle timeout.



Routing: DNS routes to Lambda when down (to trigger wake), and EC2 when up.

ACCEPTANCE CRITERIA: USER EXPERIENCE

Checklist Card

- ☐  Key entry & LocalStorage saving.
- ☐  Directory visualization (Names/Roles/Keys).
- ☐  File listing: Persistent vs. Ephemeral + TTL. 
- ☐  Inline Rendering: PDF, Markdown, Images (Sandboxed).
- ☐  Secure file upload (Encrypted, Ephemeral).
- ☐  PKI Messaging & Threading.
- ☐  Custom Branding rendering.

CRITERIA_STATUS: PENDING

UX_VERIFICATION_ID: 849-8

TIMELINE -----

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129

8

106

205

343

260

320

368

290

300

402

433

580

ACCEPTANCE CRITERIA: FLEET & INFRASTRUCTURE

Checklist Card

- ☐  Create Room (Subdomain/Repo/Region) & View Status.
- ☐  Start/Stop Logic (Boot vs. Terminate).
- ☐  GitHub → S3 Sync Trigger.
- ☐  API-only operations.
- ☐  EC2: Boot from AMI, run FastAPI, serve on 443.
- ☐  Security Critical: No Egress networking.
- ☐  Auto-shutdown on idle & DNS routing logic.

INFRA_STATUS: PENDING

FLEET_VERIFICATION_ID: 850-F

TIMELINE -----

GO

119

8

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205

348

260

320

888

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402

433

580

SUMMARY & HANDOFF

READY FOR DEVELOPMENT



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